

Machine Learning – 2 Business Report

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Business Context:

Business communities in the United States are facing high demand for human resources, but one of the constant challenges is identifying and attracting the right talent, which is perhaps the most important element in remaining competitive. Companies in the United States look for hard-working, talented, and qualified individuals both locally as well as abroad.

The Immigration and Nationality Act (INA) of the US permits foreign workers to come to the United States to work on either a temporary or permanent basis. The act also protects US workers against adverse impacts on their wages or working conditions by ensuring US employers' compliance with statutory requirements when they hire foreign workers to fill workforce shortages. The immigration programs are administered by the Office of Foreign Labor Certification (OFLC).

OFLC processes job certification applications for employers seeking to bring foreign workers into the United States and grants certifications in those cases where employers can demonstrate that there are not sufficient US workers available to perform the work at wages that meet or exceed the wage paid for the occupation in the area of intended employment.

Objective

In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labour certifications. This was a nine percent increase in the overall number of processed applications from the previous year. The process of reviewing every case is becoming a tedious task as the number of applicants is increasing every year.

The increasing number of applicants every year calls for a Machine Learning based solution that can help in shortlisting the candidates having a higher chance of VISA approval. OFLC has hired the firm EasyVisa for data-driven solutions. You, as a data scientist at EasyVisa, have to analyse the data provided and, with the help of a classification model:

Facilitate the process of visa approvals. Recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

Data Description

The data contains the different attributes of the employee and the employer. The detailed data dictionary is given below.

- `case_id`: ID of each visa application
- `continent`: Information of continent the employee
- `education_of_employee`: Information of education of the employee
- `has_job_experience`: Does the employee have any job experience? Y= Yes; N = No
- `requires_job_training`: Does the employee require any job training? Y = Yes; N = No
- `no_of_employees`: Number of employees in the employer's company
- `yr_of_estab`: Year in which the employer's company was established
- `region_of_employment`: Information of foreign worker's intended region of employment in the US.

- **prevailing_wage:** Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- **unit_of_wage:** Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
- **full_time_position:** Is the position of work full-time? Y = Full-Time Position; N = Part-Time Position
- **case status:** Flag indicating if the Visa was certified or denied

Exploratory Data Analysis

Displaying the first few rows of the dataset

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment	prevailing_wage	unit_of_wage	ful
0	EZYV01	Asia	High School	N	N	14513	2007	West	592.203	Hour	
1	EZYV02	Asia	Master's	Y	N	2412	2002	Northeast	83425.650	Year	
2	EZYV03	Asia	Bachelor's	N	Y	44444	2008	West	122996.860	Year	
3	EZYV04	Asia	Bachelor's	N	N	98	1897	West	83434.030	Year	
4	EZYV05	Africa	Master's	Y	N	1082	2005	South	149907.390	Year	

figure 1

- This Displays the first five rows of the dataset.

Displaying the last few rows of the dataset

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment	prevailing_wage	unit_of_wage	ful
25475	EZYV25476	Asia	Bachelor's	Y	Y	2601	2008	South	77092.570	Yea	
25476	EZYV25477	Asia	High School	Y	N	3274	2006	Northeast	279174.790	Yea	
25477	EZYV25478	Asia	Master's	Y	N	1121	1910	South	146298.850	Yea	
25478	EZYV25479	Asia	Master's	Y	Y	1918	1887	West	86154.770	Yea	
25479	EZYV25480	Asia	Bachelor's	Y	N	3195	1960	Midwest	70876.910	Yea	

figure 2

- This Displays the last five rows of the datasets.

Checking the data types of the columns for the dataset:

```

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   case_id                              25480 non-null  object
 1   continent                            25480 non-null  object
 2   education_of_employee                25480 non-null  object
 3   has_job_experience                   25480 non-null  object
 4   requires_job_training                25480 non-null  object
 5   no_of_employees                     25480 non-null  int64
 6   yr_of_estab                         25480 non-null  int64
 7   region_of_employment                25480 non-null  object
 8   prevailing_wage                     25480 non-null  float64
 9   unit_of_wage                        25480 non-null  object
10   full_time_position                  25480 non-null  object
11   case_status                         25480 non-null  object
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB

```

figure 3

- Most of the columns are in object types

Checking the shape of the dataset

- There are 25480 rows and 12 columns in the dataset.

Checking for duplicate values

- No Duplicate values present.

Statistical summary of the Dataset

	count	mean	std	min	25%	50%	75%	max
no_of_employees	25480.000	5667.043	22877.929	-26.000	1022.000	2109.000	3504.000	602069.000
yr_of_estab	25480.000	1979.410	42.367	1800.000	1976.000	1997.000	2005.000	2016.000
prevailing_wage	25480.000	74455.815	52815.942	2.137	34015.480	70308.210	107735.513	319210.270

figure 4

Observation

- The average number of employees is approximately 5,667, while the median is much lower
- This large difference between mean and median indicates a highly right-skewed distribution.
- The presence of a negative minimum value (-26) highlights data quality issues that need correction.
- The mean establishment year is around 1979, with most companies established between 1976 and 2005.
- The minimum year (1800) represents very old firms and may act as outliers, though they are valid.

Univariate Analysis

The dataset consists of visa applications processed by OFLC, containing both categorical and numerical variables related to the applicant, employer, and job characteristics. Categorical variables such as continent, education level, job experience, job training requirement, region of employment, wage unit, and full-time position show that most applications originate from Asia, involve Bachelor's or Master's degree holders, are full-time positions, and often require prior job experience. The target variable (case_status) is imbalanced, with a higher proportion of Certified cases compared to Denied, indicating that most applications are approved.

Univariate Analysis on numerical column

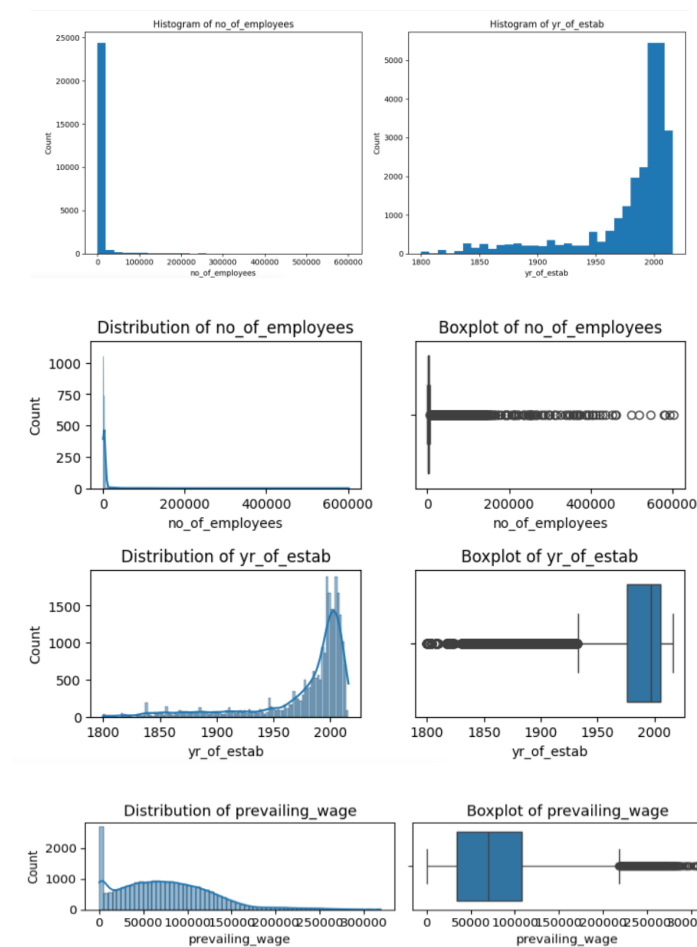


figure 5

Observation

- The distribution is highly right-skewed.
- The boxplot shows many outliers, indicating large variability.
- Here variable are not normally distributed and may need outlier treatment.
- year of est is highly left-skewed.
- The boxplot shows many outliers, indicating large variability.
- Prevailing age has highly right-skewed.

Univariate Analysis for Categorical variable

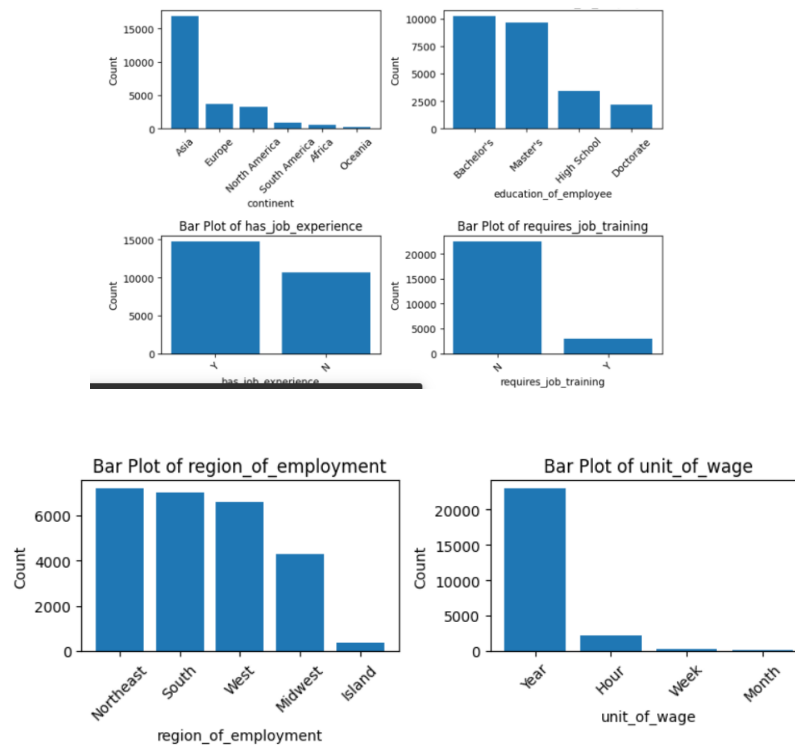


figure 6

Observation

- **Continents:** Most applicants are from Asia, far more than any other continent. Europe and North America contribute moderate numbers. Africa, South America, and especially Oceania have very few applications. This shows the dataset is highly skewed toward Asian applicants.
- **Education of employee:** Majority of applicants have a Bachelor's or Master's degree. High School and Doctorate categories are much smaller. This indicates most visa applicants are well-educated professionals.
- **Has Job Experience:** More applicants have prior job experience (Y) than not. Still, a significant portion has no experience, so both groups are important.
- **Requires Job Training:** Vast majority of jobs do not require training (N). Very few jobs require training (Y), showing employers prefer ready-to-work candidates.
- **Region of Employment:** Most jobs are in Northeast, South, and West regions. Midwest has fewer jobs compared to these. Island region has very few job opportunities.
- **Unit of Wage:** Most wages are reported on a Yearly basis. Hourly wages are much fewer. Weekly and Monthly wages are very rare. So, salary is mostly expressed as **annual pay**.

Bivariate analysis Continent vs case status

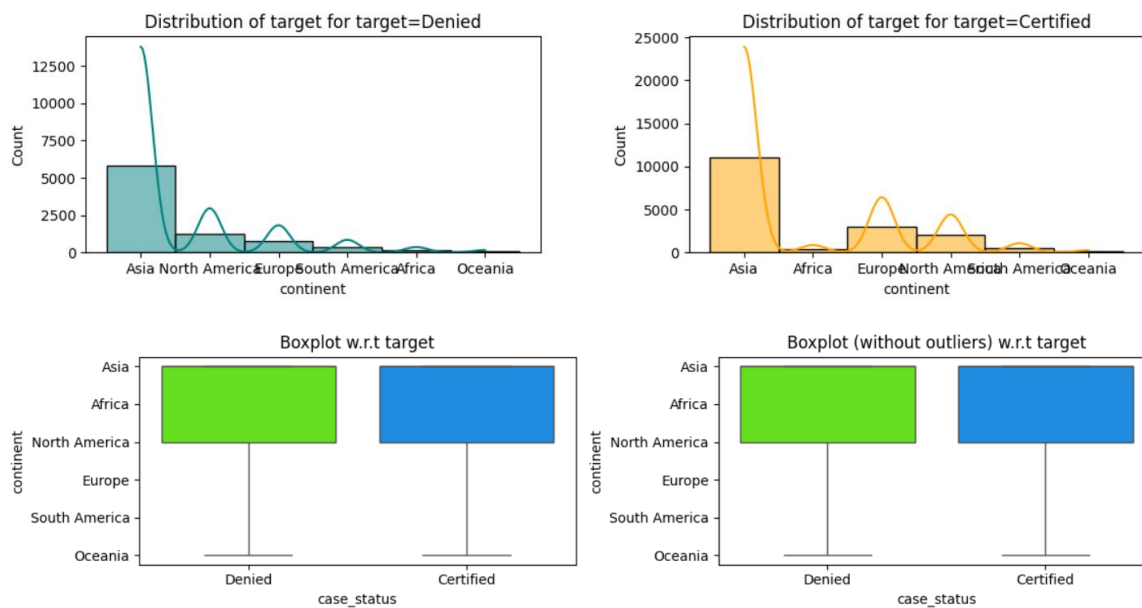
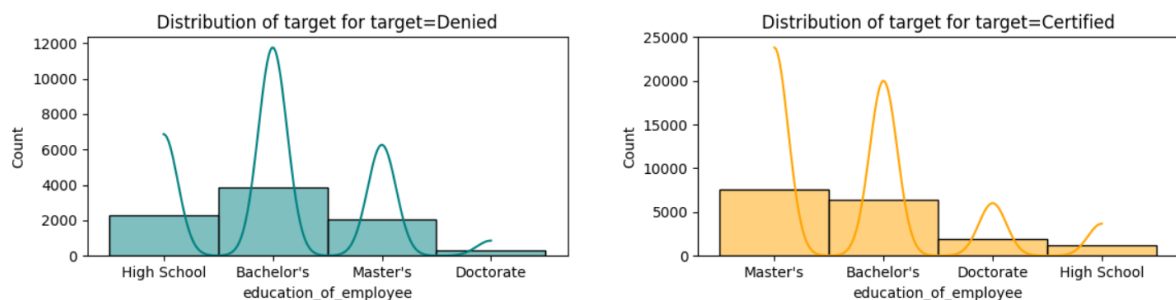


figure 7

Observation

- Asia shows higher number of denial across all the continents.
- north America and Europe show moderate denial.
- Africa, South America, and Oceania have very few denied cases.
- Asia has slightly higher number of certified
- Europe and north America have substantial certified counts.
- Overall, certifications are significantly higher than denials across all continents

Bivariate Analysis Education of Employee vs case status



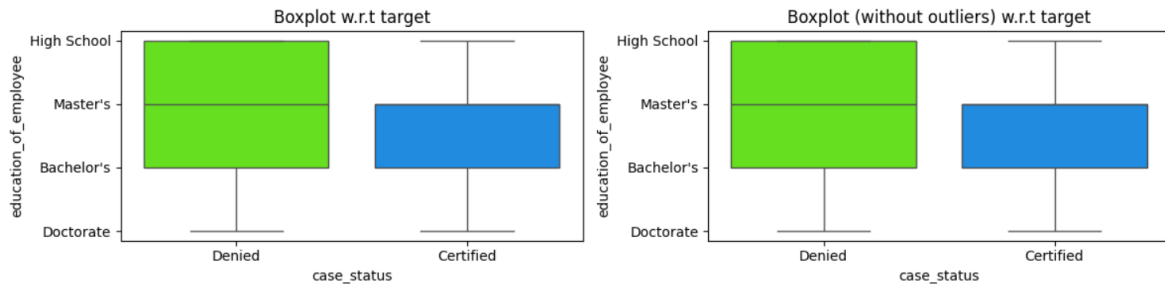


figure 8

Observation

- Higher number of denial application is from candidate holding bachelor's degree.
- Master degree holder has noticeable number of denial but fewer than bachelor degree holder.
- Doctorate holder has very fewer denial.
- Master's degree holders have the highest number of certified visas.
- Bachelor's degree holders also show strong certification counts
- Doctorate candidates have fewer applications but a relatively higher success rate.
- High School candidates have the lowest number of certifications.

Bivariate Analysis of Has Job Experience vs case status

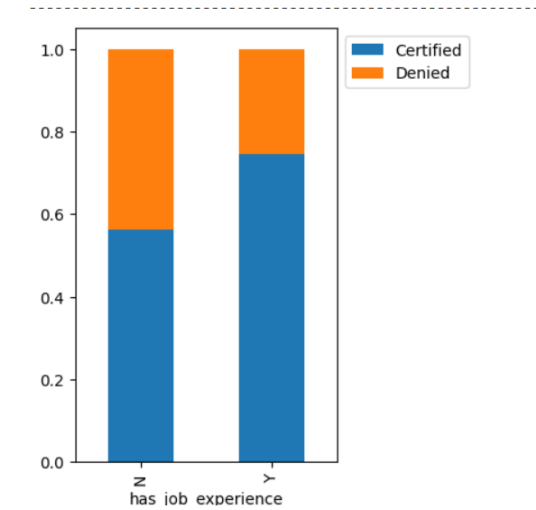


figure 9

Observation

- Applicants with prior job experience have a much higher proportion of certified visas compared to denials.
- Applicant with no job experience has more denial proportion compare to with experience.
- certification rate is slightly higher compare to inexperienced applicants

- This indicates that the job experience influences the visa approvals

Bivariate Analysis on Required Job training vs case status

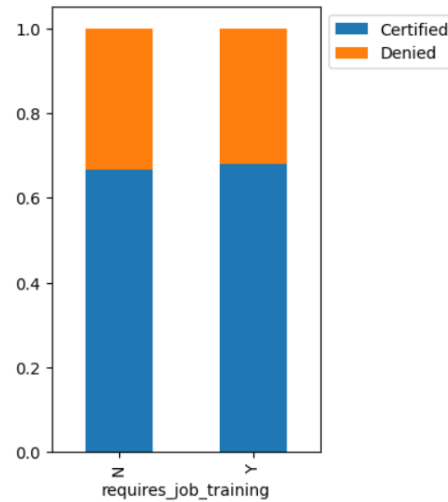


figure 10

Observation

- The proportion of certified visa is slightly higher compare to denials who require Job training.
- candidate who requires training is slightly higher.
- Both categories show that certifications are significantly higher than denials overall.

Bivariate Analysis on Region of employment vs case status

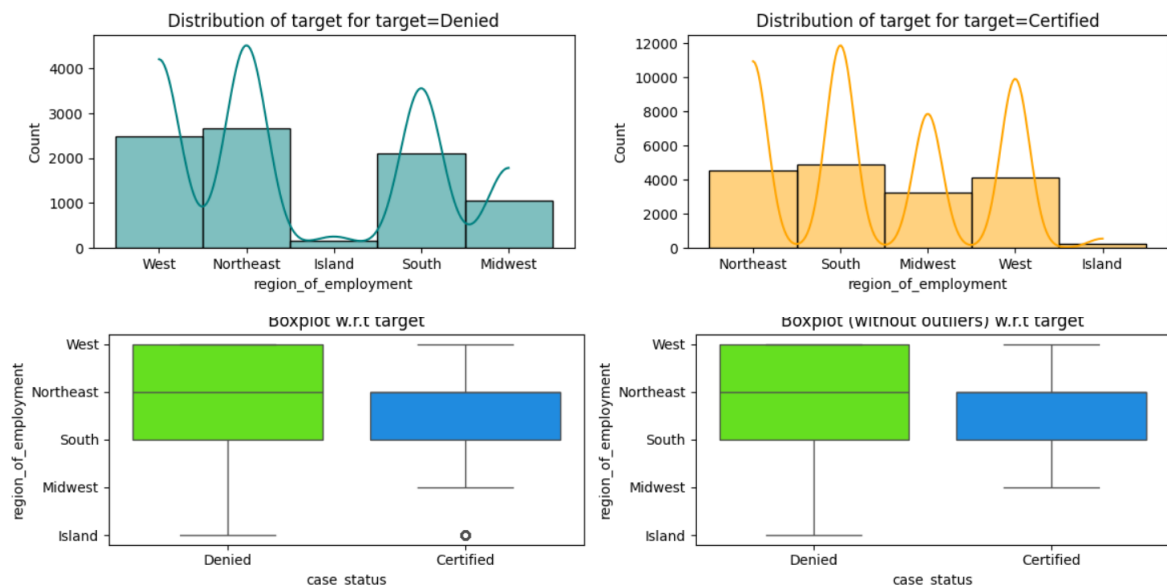


figure 11

Observation

- Most applications (both denied and certified) come from West, South, and Northeast regions.
- Island region has very few cases in both categories
- Northeast and West have relatively higher counts for denial.
- South also contributes a significant number of denials.
- For certified West and South dominate, followed by Northeast.
- Midwest has moderate representation. For certified counts.
- West and South regions show more certifications than denials - higher approval tendency.
- Northeast has a noticeable share in both denied and certified, indicating mixed outcomes.
- Region of employment is an important driver for visa outcome.
- Jobs in West and South regions have higher chances of certification.
- Applications from Island and some Midwest areas are more likely to be denied or have fewer successful cases.

Bivariate Analysis on No of Employee vs case status

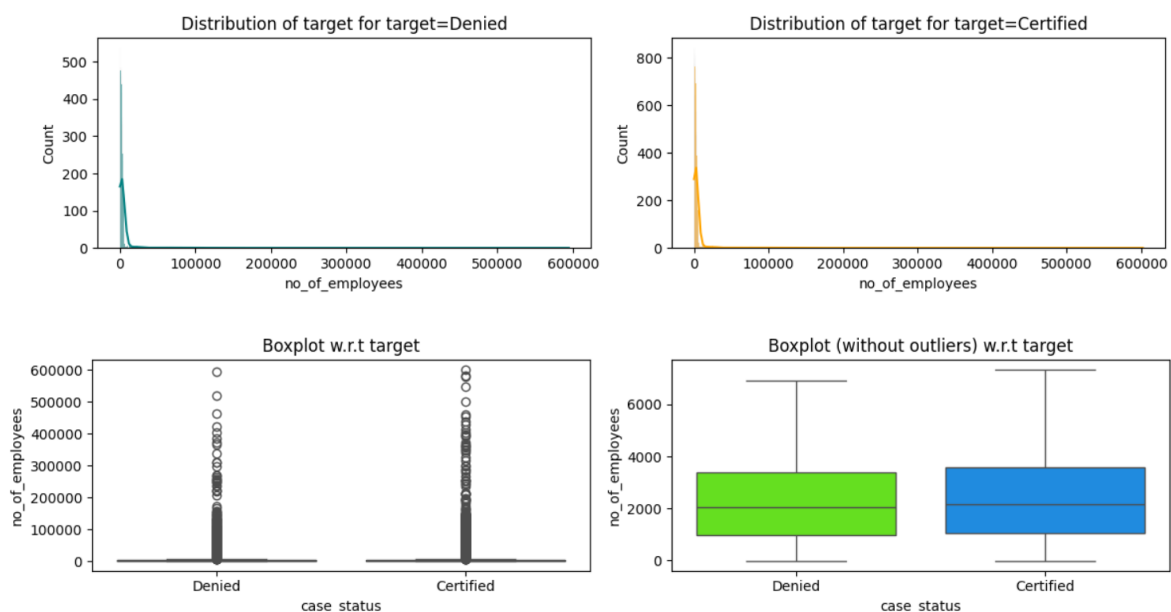


figure 12

Observation

- Distribution is highly right skewed.
- most denial are comes from companies with small number of employees.
- This indicate that the smaller organisation account for visa denials.
- Certification counts are generally higher than denials across all company sizes.
- no_of_employees does not follow a normal distribution and contains extreme outliers.
- Both certified and denied applications are dominated by small companies, likely due to higher volume.

Bivariate analysis on full time position vs case status

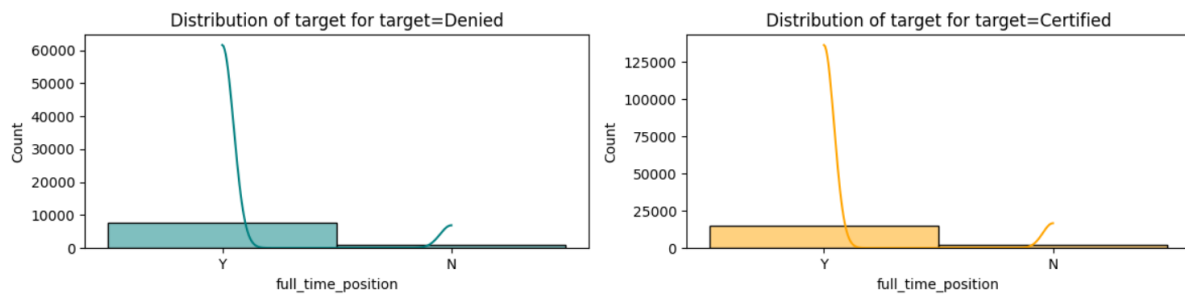


figure 13

Observation

- majority of the application are denials for full time compare to non-full time.
- majority of the application are certified for full time compare to non full time.
- full time position dominates visa approvals over all.
- Full-time roles have a much higher likelihood of certification compared to part-time or non-full-time roles.

Bivariate Analysis on year of establish vs case status

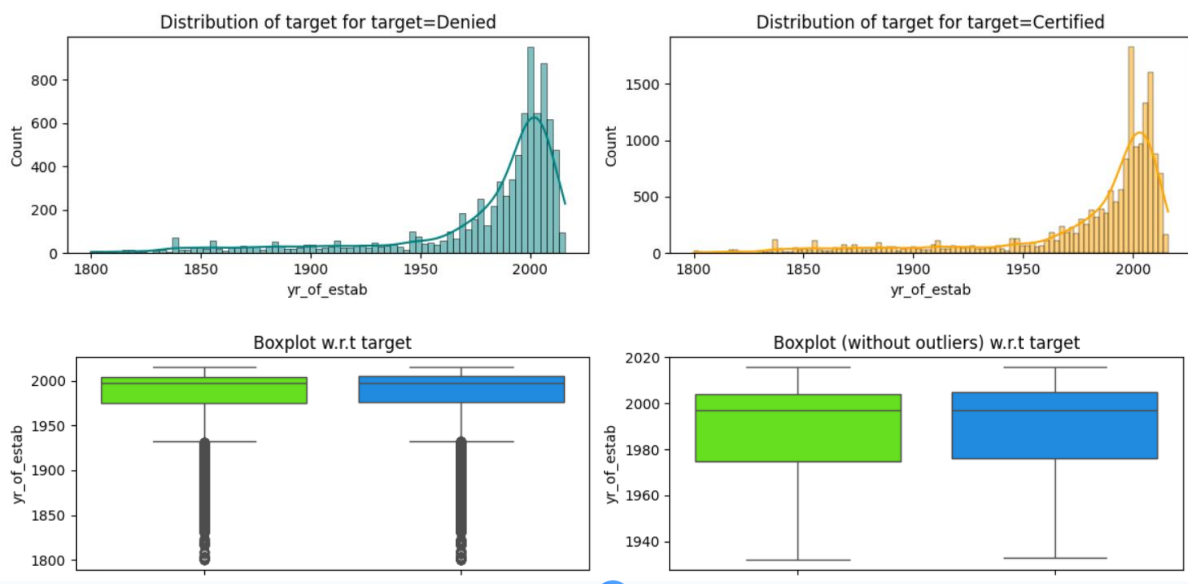


figure 14

Observation

- Both denied and certified cases are heavily concentrated around companies established after 1950
- Very old companies (before 1900) are rare in the dataset for both outcomes.
- The shape is right-skewed: more recent establishments dominate.
- Certified cases are slightly more concentrated in newer companies compared to denied cases.

- Denied cases show a wider spread across older establishment years, including more very old firms.
- Median year for both groups is around 1995–2000.
- Certified cases have a slightly higher median → newer firms are marginally more likely to get certification.
- Year of establishment has a mild influence on visa outcome.
- Newer companies have a slightly higher chance of certification.
- However, the overlap is large, so this feature alone cannot strongly separate certified vs denied cases.

Bivariate Analysis on unit on wage vs case status

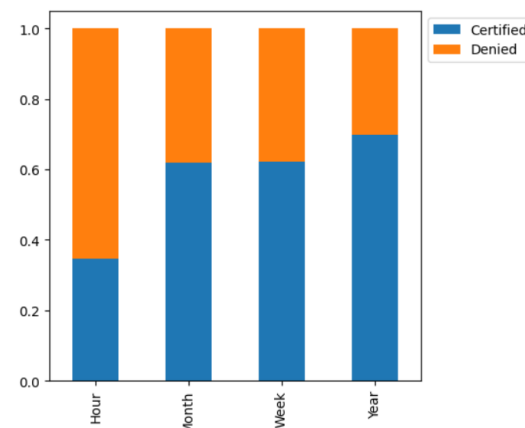


figure 15

Observation

- Hourly wages have the highest denial rate: Only about 35% certified and 65% denied. Hour-based jobs are more likely to be rejected.
- Monthly and Weekly wages show better approval: Around 62–63% certified. Denials are much lower compared to hourly jobs.
- Yearly wages have the best outcome: About 70% certified and only 30% denied. Jobs offering annual salaries are most likely to get certification.
- Wage structure strongly affects visa outcome.
- Applications with yearly salaries have the highest chance of certification.
- Hourly wage jobs are at highest risk of denial.
- This variable is an important predictor in your model.

Bivariate Analysis on prevailing wage vs case status

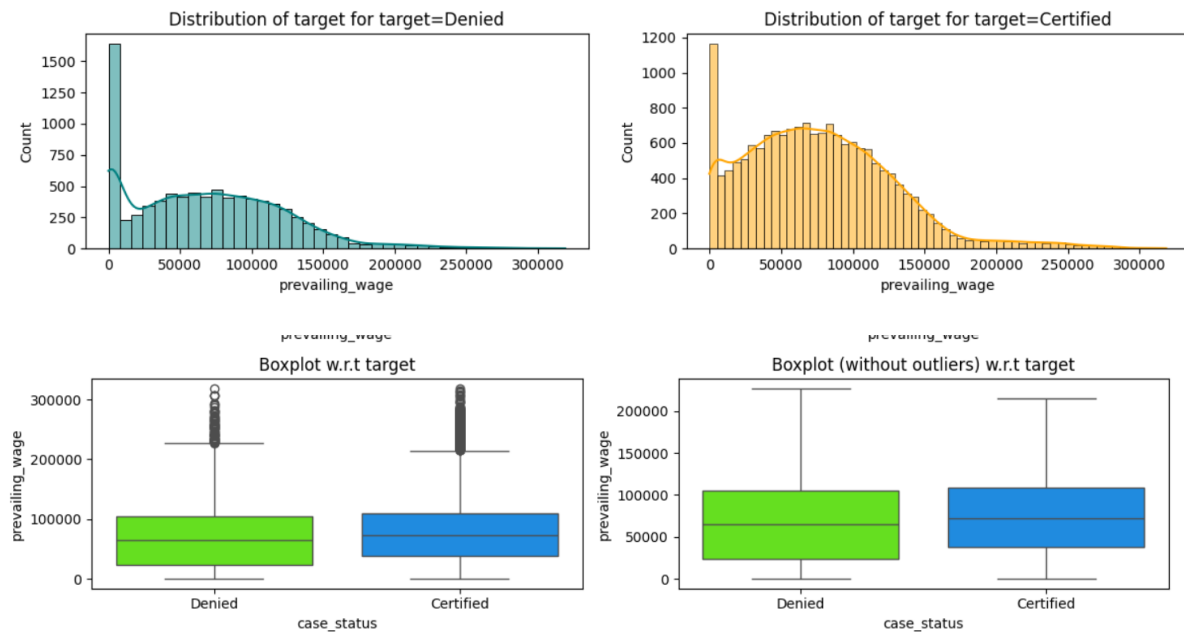


figure 16

Observation

- Both denied and certified wages are right-skewed (more cases at lower–mid wages, few at very high wages).
- Certified cases are more concentrated in the higher wage ranges compared to denied cases.
- Denied cases show many applications at very low wages, especially near zero.
- Comparison between Denied and Certified: Certified applications generally have higher prevailing wages than denied ones. Denied cases are more common in the lower wage bands.
- In Box plot Median wage for Certified is greater than Denied.
- Certified group has a higher upper quartile and higher spread.
- Many outliers exist on the higher end for both, but more extreme values appear in certified.
- Prevailing wage is a strong driver of visa outcome.
- Higher-paying jobs have a much better chance of being certified.
- Low-wage jobs are more likely to be denied

Data Preprocessing

Outlier Detection

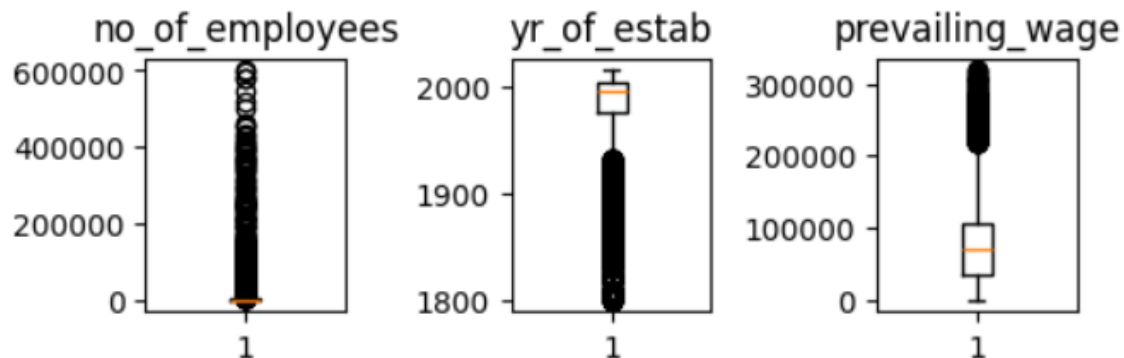


figure 17

Observation

- There are large number of outliers are present in the dataset.
- These are genuine outliers. Hence not Treating

Data preparing for modelling

- Converted the case status into numeric using map function.
- Defined dependent and target variable
- Split the data into three parts training validation and testing.
- The value of Xtrain rows and column is 15288, 10
- The value of Xval row and columns is 5096, 10
- The value of Xtest rows and columns are 5096, 10

Feature Engineering

- No Duplicate values present
- No Missing values present in the dataset
- Case status is in categorical form converted into numerical using map function.
- There are negative value present in the no of employment. So it is treated after splitting the data into train validation and testing using simple Imputer median
- Created dummy variable for train validation and test data using one hot encoding method

Model Building

Model evaluation criterion

Model evaluation criteria are the metrics used to measure how well a machine learning model is performing, especially on unseen (validation or test) data.

Accuracy- Measures the overall correctness of the model. It tells what percentage of total predictions are correct. Not reliable when data is imbalanced like more Certified than Denied cases

Recall: Measures how many actual positive cases are correctly identified. recall is very important because missing a genuine “Certified” case (predicting it as Denied) is costly. High recall means fewer false negatives.

Precision- Measures how many predicted positives are actually correct. High precision means when the model predicts Certified it is usually right. Important to avoid wrongly approving weak profiles.

F1-score: mean of Precision and Recall. Useful when data is imbalanced and both false positives and false negatives matter. A single balanced metric to compare models.

Here in Easyvisa dataset we will focus on f1 score.

Build 5 models (from decision trees, bagging and boosting methods)

Model Building - Original Data

Build five model Bagging, Random forest, GBM, AdaBoost and Decision tree with original Data below is the f1score for train and validation performance of original Data

```
..
  Training Performance:

  Bagging: 0.9892146288851849
  Random forest: 1.0
  GBM: 0.8289152182933014
  Adaboost: 0.8199619254827305
  dtree: 1.0

  Validation Performance:

  Bagging: 0.7773232028053769
  Random forest: 0.804946599213041
  GBM: 0.8256702319766634
  Adaboost: 0.8185170012290045
  dtree: 0.7444117647058823
```

figure 18

Observation

- Decision Tree and Random Forest show perfect training scores (1.0), which means they fit the training data extremely well.
- But their validation scores drop (Decision Tree - 0.744, Random Forest -0.805), indicating overfitting—they memorize training data but generalize less well.
- Bagging has very high training performance (0.989) but lower validation performance (0.777).
- This also suggests overfitting, though less severe than Decision Tree.
- GBM and AdaBoost have lower training scores.
- This shows better generalization and more stable learning.
- Among all models, GBM and AdaBoost have the smallest gap between training and validation, meaning they balance bias and variance better.

Insight:

- Overall, GBM and AdaBoost are the most reliable models, while Decision Tree and Random Forest need regularization or hyperparameter tuning to reduce overfitting.

Model Comparison on training and validation data of original data

*** Model Performance Comparison:

	Model	Train F1	Validation F1
0	Bagging	0.989	0.777
1	Random Forest	1.000	0.805
2	GBM	0.829	0.826
3	AdaBoost	0.820	0.819
4	Decision Tree	1.000	0.744

figure 19

Training and Validation Performance Difference:

	Model	Training f1	Validation f1	Difference
0	Bagging	0.989	0.777	0.212
1	Random Forest	1.000	0.805	0.195
2	GBM	0.829	0.826	0.003
3	AdaBoost	0.820	0.819	0.001
4	Decision Tree	1.000	0.744	0.256

figure 20

Observation

- Decision Tree and Random Forest have perfect training F1 (1.000), which clearly indicates overfitting. Their validation F1 drops to 0.744 (Decision Tree) and 0.805 (Random Forest), meaning they do not generalize well.
- Bagging also shows very high training F1 (0.989) but a much lower validation F1 (0.777). This still indicates overfitting, though less extreme than a single decision tree.
- GBM (Gradient Boosting) shows almost equal training and validation F1 (0.829 vs 0.826). This indicates good generalization and a well-balanced model.

- AdaBoost also has very similar training and validation F1 (0.820 vs 0.819), showing stable performance
- Overall, GBM is the best model here because: It has the highest validation F1 (0.826),
- And the smallest gap between training and validation, meaning low overfitting

Insight:

- Overall, GBM is the best model followed by AdaBoost because: It has the highest validation F1

Model Building - Oversampled Data

Oversampling is a technique used to handle class imbalance in a dataset.

When one class has many more samples than the other, the model becomes biased toward the majority class. Oversampling fixes this by increasing the number of samples in the minority class.

Why we are doing oversampling

Certified cases are majority Denied cases are minority so Data is imbalance.

If trained directly: Model may predict mostly Certified Poor performance on Denied cases so Oversampling balances this.

How Oversampling Works, It adds more minority-class samples by: Copying existing minority samples (simple oversampling) Creating synthetic samples (SMOTE)

In simple term Oversampling means: Creating more samples for the less frequent class so that both classes are balanced and the model learns.

We have built 5 model using oversample technique below are the model performance.

Model Performance (Oversampled Data):

	Model	Train F1 (Oversampled)	Validation F1
0	Bagging	0.987	0.765
1	Random forest	1.000	0.799
2	GBM	0.805	0.815
3	Adaboost	0.796	0.813
4	dtree	1.000	0.726

figure 21

Observation

- Decision Tree and Random Forest again show perfect training F1 (1.000), which means they are overfitting heavily. Their validation F1 drops to 0.726 decision tree and 0.799 (Random Forest), so generalization is weak.
- Bagging has very high training F1 (0.987) but much lower validation F1 (0.765), showing overfitting, though less extreme than a single tree.
- GBM (Gradient Boosting) shows balanced behaviour. Validation is slightly better than training - indicates good generalization.
- AdaBoost also shows stable performance: Very small gap - low overfitting

Insight

- Best model on oversampled data: GBM has the highest validation F1 (0.815), closely followed by AdaBoost (0.813).
- GBM is the best choice because it combines high validation performance with strong generalization.

Model Building - Under sampled Data

Under sampling is a technique used to handle class imbalance by reducing the number of samples in the majority class. Instead of increasing minority data (like oversampling), under sampling removes some majority data to make both classes balanced.

Why Under sampling is Needed:

In our dataset Certified is very large Denied is small

If trained directly: Model becomes biased toward Certified Under sampling fixes this by cutting down Certified samples. We have built 5 model using under sample technique

... Model Performance (Undersampled Data):

	Model	Train F1 (Undersampled)	Validation F1
0	Bagging	0.981	0.704
1	Random Forest	1.000	0.743
2	GBM	0.727	0.777
3	AdaBoost	0.702	0.766
4	Decision Tree	1.000	0.699

figure 22

Observation

- Decision Tree and Random Forest have perfect training F1 (1.000), which clearly indicates severe overfitting. Their validation F1 drops to 0.699 and 0.743 respectively, so they do not generalize well
- Bagging also shows overfitting: large gap, poor generalization.
- GBM (Gradient Boosting) shows balanced behaviour: Train F1 is 0.727, Validation F1 is 0.777 Validation is higher than training- indicates good generalization.

- AdaBoost also generalizes well: Train F1 is 0.702, Validation F1 is 0.766 Small gap stable performance.
- Best model on under sampled data: GBM has the highest validation F1 (0.777), followed by AdaBoost (0.766). Tree-based single models and bagging are not suitable here due to heavy overfitting.
- Boosting models (GBM, AdaBoost) handle under sampled data better.

Insight

- Boosting models (GBM, AdaBoost) handle under sampled data better.

F1 score difference of training and validation score of undersample data

... Training and Validation F1 Score Difference:

	Model	Training F1	Validation F1	Difference (Train - Val)
0	Bagging	0.981	0.704	0.277
1	Random Forest	1.000	0.743	0.257
2	GBM	0.727	0.777	-0.050
3	AdaBoost	0.702	0.766	-0.065
4	Decision Tree	1.000	0.699	0.301

figure 23

Observation

- Decision Tree: There is a Very large gap (0.301) - Strong overfitting.
- Bagging: Big gap (0.277) - Overfitting, though better than single tree.
- Random Forest: big gap Still overfitting, but validation performance is better than Bagging and Decision Tree.
- GBM: Negative gap (-0.050) → Good generalization, not overfitting.
- Validation performance is even better than training.
- Negative gap (-0.065) - Stable model, good generalization.
- AdaBoost: Stable model, good generalization.

Hyperparameter Tuning

Hyperparameter tuning is the process of finding the best values for model settings (called hyperparameters) so that the model gives the best performance. Hyperparameters are values that you set before training the model. They are not learned from data.

Decision Tree: max_depth, min_samples_split

Random Forest: n_estimators, max_features

Gradient Boosting: learning_rate, n_estimators

AdaBoost: `n_estimators`, `learning_rate`

These are the hyper parameter Why hyper parameter are used:

- Wrong values can cause overfitting (model memorizes data)
- Or underfitting (model is too simple)
- Good tuning helps the model generalize well on new data

It is done by trying many combinations of hyperparameters and choose the one with best performance. Common methods:

- Grid Search – tries all combinations
- Random Search – tries random combinations

Hyperparameter tuning means: Trying different model settings and selecting the one that gives the best result on validation data.

Tuning AdaBoostClassifier model with oversample data

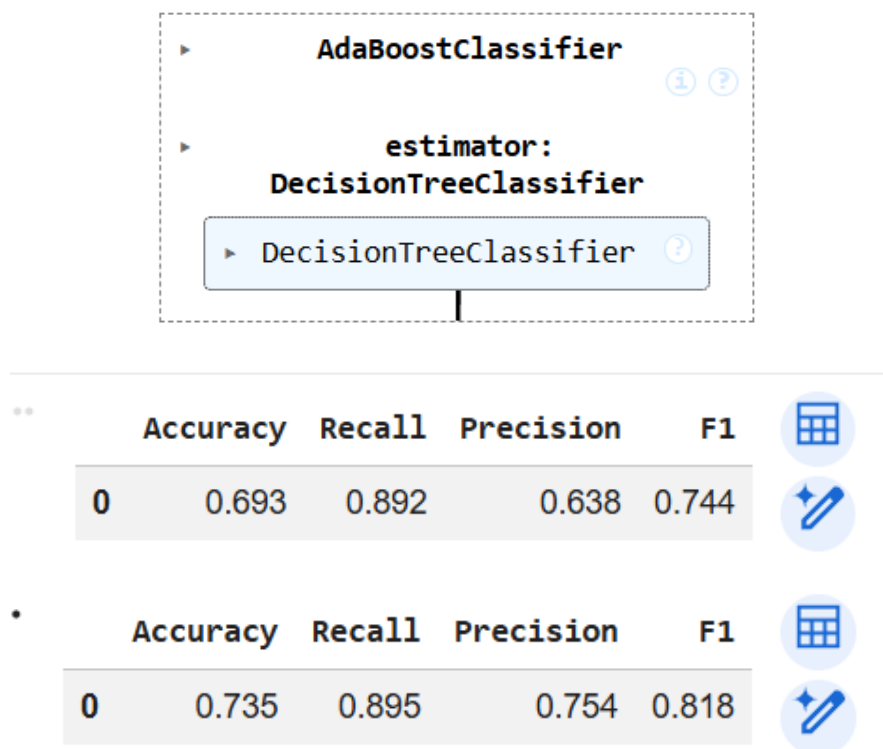


figure 24

Observation

- The tuned AdaBoost model shows strong recall (~0.89–0.90) in both evaluations, meaning it is very good at correctly identifying positive cases (e.g., Certified visas). This is important for your business goal of not missing good candidates.
- After tuning, the model's overall performance improved:
- The increase in precision means fewer false approvals, while still keeping high recall. This shows that hyperparameter tuning helped balance the trade-off between catching positives and avoiding wrong positives.

- The higher F1 score (0.818) indicates better overall balance between precision and recall compared to the earlier version.

Insight:

- Hyper-tuning AdaBoost on oversampled data made the model: More accurate More reliable in approvals
- This tuned AdaBoost is a strong for final model selection.

Tuning Gradient Boosting model with oversampled Data

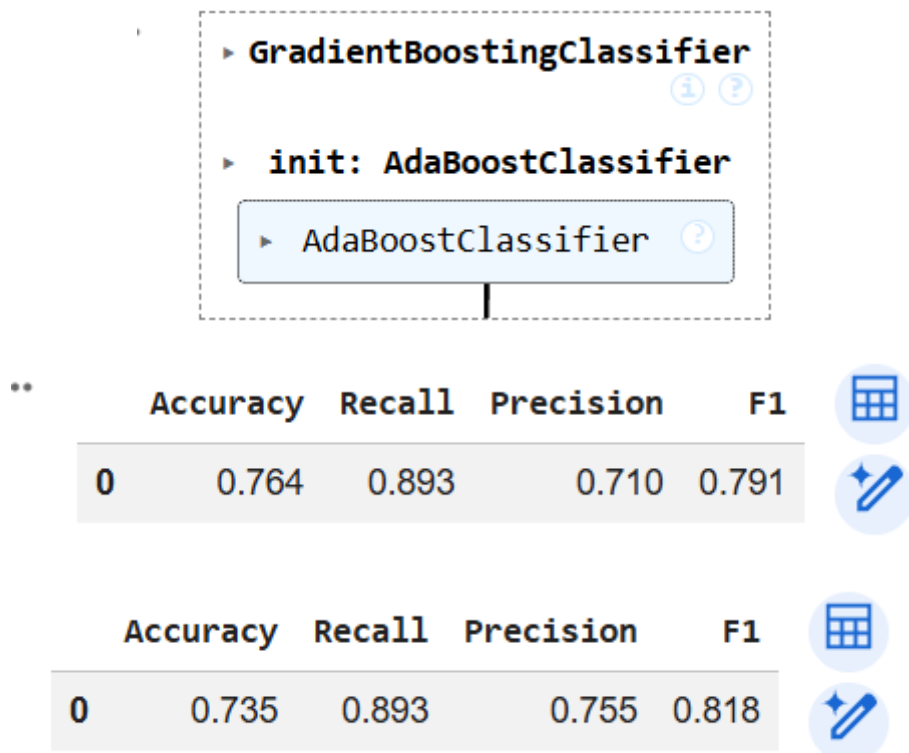


figure 25

Observation:

- Recall is exactly the same (0.893) for both training and validation, which shows the model is very stable in identifying positive cases and is not overfitting on recall.
- Validation precision (0.755) is higher than training precision (0.710). This means the model is making fewer false positives on unseen data, which is a good sign.
- Validation F1 score (0.818) is higher than training F1 (0.791). This indicates good generalization and no overfitting.
- Accuracy drops slightly from 0.764 to 0.735, which is normal and acceptable when moving from training to validation.

Insight:

- The tuned Gradient Boosting model generalizes well, shows no overfitting, and performs even better on validation in terms of precision and F1. It is a strong and reliable model for final selection.

Tunning Gradient Boosting model using under sample technique

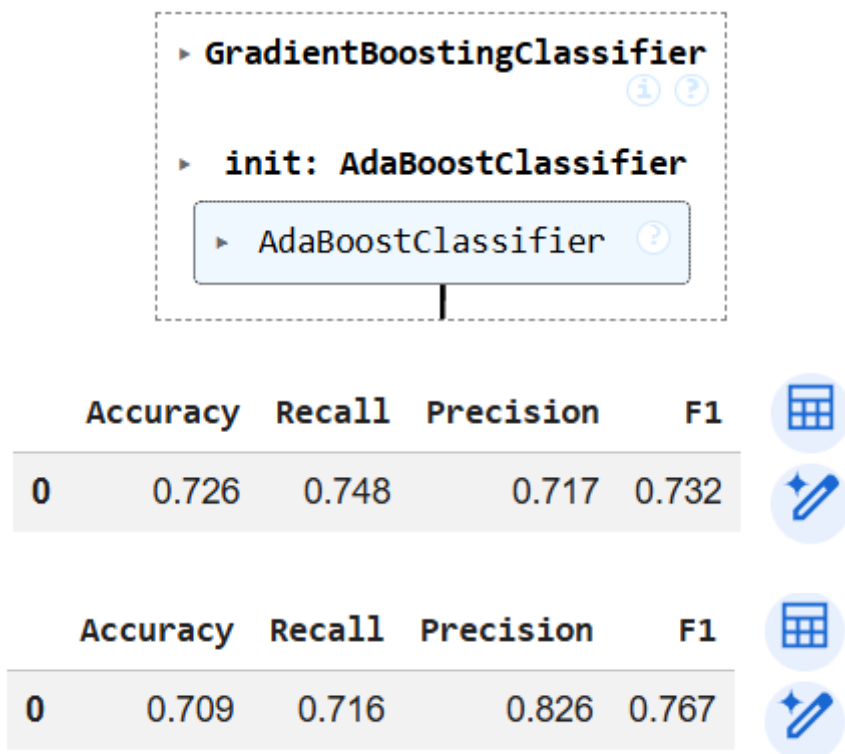


figure 26

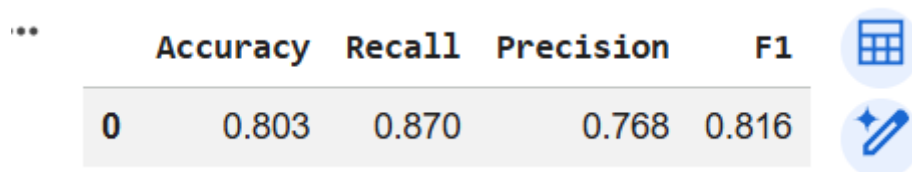
Observation

- Accuracy drops slightly from training (0.726) to validation (0.709), which is normal and shows no major overfitting.
- Recall decreases from 0.748 to 0.716, meaning the model misses slightly more positive cases on unseen data.
- Precision increases a lot from 0.717 (train) to 0.826 (validation). This means on validation data, when the model predicts “positive,” it is more often correct.
- F1 score improves from 0.732 to 0.767, showing that overall balance between precision and recall is better on validation than training.

Insight:

- This tuned Gradient Boosting model generalizes well. It does not show overfitting, and its validation performance—especially precision and F1—is better than training, making it a stable and reliable model.

Tunning Random Forest model using over sample



	Accuracy	Recall	Precision	F1
0	0.755	0.858	0.792	0.824

figure 27

Observation

- There is a small drop in accuracy from training (0.803) to validation (0.755), which is expected and indicates only mild overfitting.
- Recall remains high and stable (0.870 $\rightarrow$ 0.858). This is very good for your problem since catching positive cases (e.g., certified visas) is important.
- Precision improves on validation (0.768 $\rightarrow$ 0.792), meaning the model makes fewer false positive predictions on unseen data.
- F1 score slightly increases from 0.816 to 0.824, showing better overall balance between precision and recall on validation.

Insight

- Random Forest with oversampling generalizes well. It maintains high recall, improves precision on validation, and shows no serious overfitting. This makes it one of the strongest and most reliable models among your tuned models.

Model Comparison and Final Model Selection

Training performance comparison:

	Tuned Gradient Boosting (Undersampled)	Tuned Gradient Boosting (Oversampled)	Tuned AdaBoost (Oversampled)	Tuned Random Forest (Oversampled)
Accuracy	0.726	0.764	0.693	0.803
Recall	0.748	0.893	0.892	0.870
Precision	0.717	0.710	0.638	0.768
F1	0.732	0.791	0.744	0.816

xt steps: [Generate code with models, train, cross_val](#) [New interactive sheet](#)

figure 28

Observation

- Best overall training performance: Tuned Random Forest (Oversampled) is the strongest on training data. This shows it learns patterns very well from the oversampled data.
- Recall: Tuned Gradient Boosting (Oversampled) and Tuned AdaBoost (Oversampled) both give very high recall. This is useful when missing positive cases (e.g., certified visas) is costly.
- Precision: AdaBoost (Oversampled) has the lowest precision (0.638), meaning it produces more false positives. Random Forest has the best balance between recall and precision.

Test Performance comparison

Test performance comparison:

	Tuned Gradient Boosting (Undersampled)	Tuned Gradient Boosting (Oversampled)	Tuned AdaBoost (Oversampled)	Tuned Random Forest (Oversampled)
Accuracy	0.695	0.722	0.721	0.732
Recall	0.718	0.890	0.889	0.848
Precision	0.804	0.744	0.743	0.773
F1	0.759	0.810	0.810	0.808

figure 29

Observation

- Tuned Gradient Boosting (Oversampled) and Tuned AdaBoost (Oversampled) both achieve the highest F1 is 0.810.
- This means they give the best balance between precision and recall on unseen data.
- Tuned Gradient Boosting (Oversampled) has the highest recall of 0.890, followed very closely by AdaBoost (0.889).
- So Gradient Boosting (Oversampled) is best at correctly identifying “Certified” cases.
- Tuned Random Forest (Oversampled) has the highest accuracy = 0.732, but its recall (0.848) and F1 (0.808) are slightly lower than GBM and AdaBoost.
- Tuned Gradient Boosting (Undersampled) performs worst overall (Accuracy = 0.695, F1 = 0.759). This shows that undersampling removed too much useful information.

Insight

- Tuned Gradient Boosting (Oversampled) is the overall best model.

Actionable Insights

- Applicants with higher prevailing wages have a much higher chance of getting certified. Denied cases are concentrated at lower wage ranges, while certified cases shift toward mid-to-high wage levels.
- Applications with **Yearly wages** show much higher certification rates than Hourly or Weekly wages. Hourly wage cases have the highest denial proportion.
- Regions like **West, South, and Northeast** have higher volumes of certified cases compared to Midwest and Island, which show relatively lower certification proportions.
- Applicants with **Bachelor’s or Master’s degrees** and those having **prior job experience** are more likely to be certified than those with only High School education or no experience.
- Employers with more employees and more recent establishment years tend to have higher certification rates, indicating stronger compliance and stability.
- Models trained with oversampled data consistently outperform undersampled ones, especially in recall and F1-score, which are critical for identifying likely certified cases.
- Best model for deployment Tuned Gradient Boosting (Oversampled) gave the best trade-off with Recall approx. 0.89 strong f1 0.81
- This is ideal for minimizing false rejections of good candidates.

Actionable Business Recommendations

- Prioritize high-probability profiles

- Fast-track applications with: Higher prevailing wages, Yearly wage unit, Bachelor's/Master's degree, Prior job experience, and Employers with larger workforce
- Guide employers on salary strategy: Advise employers to align offered wages with or above regional industry standards to improve certification chances.
- Strengthen weak applications: Suggest skill upgrades or certifications, Recommend offering training if experience is lacking, Adjust job roles to better match prevailing wage benchmarks
- Focus marketing and employer outreach more in high-success regions (West, South, Northeast) while designing special support programs for low-success regions.
- Use ML model for pre-screening: Deploy the tuned Gradient Boosting model as a decision-support tool to: Flag high-chance candidates early, reduce manual review workload, Improve overall processing efficiency
- Continuous monitoring and retraining Retrain the model periodically with new data to adapt to policy changes, labour market shifts, and wage trends.
- Business impact Using this system can: Reduce processing time, Improve certification success rates, Help employers make better hiring offers